

Tutorial: A Parametric Finger-Jointed Box

Build a six-panel finger-jointed box from scratch in the ParamWeave graph. Eight constants drive everything (outer size, material thickness, laser kerf and finger counts), so changing one number regenerates the whole box.

Each panel ends up as an ordinary FreeCAD sketch plus an extruded solid. The sketches are ready to export for laser cutting.

What you will use: Number, Expression, Finger Joint Panel, Sketch and Extrude nodes; wiring by clicking ports; the property panel; Duplicate; Evaluate.

Time: about 20 minutes. **Result:** 33 nodes, 79 wires.

Finished file: `finger-jointed-box.FCStd` (next to this tutorial) is the result of following every step. Open it in FreeCAD to compare with your own graph or to skip ahead. Its six sketches and panels are ordinary FreeCAD objects, so they display even without ParamWeave installed. With ParamWeave, the graph pane shows the full graph and **Evaluate** rebuilds the box.

Every step in this tutorial was performed through the GUI and checked automatically: `python3 tools/run_freecad_tests.py --gui-script tests/gui/tutorial_finger_box.py` rebuilds the box this way and regenerates these pictures and the finished `.FCStd` file.

Before you start: working in the graph

- **Add a node:** right-click an empty spot on the graph canvas, open a category and pick the node. It appears where you clicked.
- **Select a node:** click its title bar. The property panel below the graph shows its settings. Ctrl-click (⌘-click on macOS) adds more nodes to the selection, or drag a box around them.
- **Edit a setting:** click the field in the property panel, type, and press Return.
- **Rename a node:** edit its **Label** field. Labels are only names; renaming never breaks wires.
- **Wire two nodes:** click an *output* port dot (right side of a node), then an *input* port dot (left side). Esc cancels a half-made wire. Wiring into an input that already has a wire replaces the old wire.
- **Wired settings:** every number setting has an input port with the same name. Once wired, the field shows *driven by input* and the wire's value is used.
- **Delete:** select a node or click a wire, then press Delete.
- **Duplicate:** select nodes and press Ctrl+D (⌘D on macOS), or right-click the empty canvas and choose **Duplicate Selected**. Copies keep the wires coming into them.
- **Move and view:** drag a node by its title bar; scroll to zoom; press F to frame everything.
- **Evaluate:** click **Evaluate** at the top of the graph pane to build or update the geometry.

How the box fits together

The box is $160 \times 100 \times 70$ mm made of 3 mm material, with panels at their full outer size. Along every joint one panel has *fingers* and the other has matching *slots*. Each panel edge has a **mode**:

- **out**: starts and ends with a finger at the panel's outer edge.
- **in**: starts and ends with a slot, cut in by the material thickness.

Two edges that meet always use opposite modes and the same finger count. Finger counts must be odd so every edge reads the same from either end.

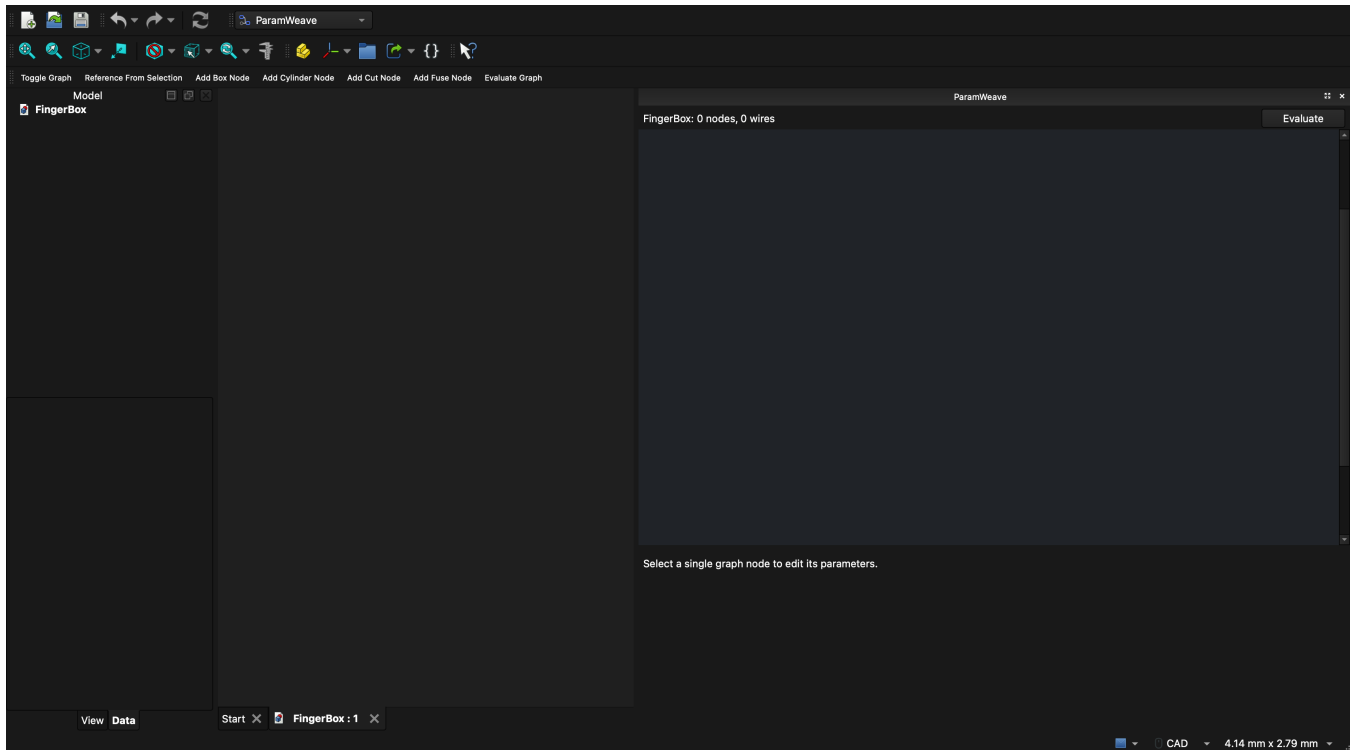
Panel	Size	Bottom / top edges	Left / right edges	Plane	Offset
Bottom	$L \times W$	in, L fingers	in, W fingers	XY	0
Lid	$L \times W$	in, L fingers	in, W fingers	XY	$H - t$
Front	$L \times H$	out, L fingers	out, H fingers	XZ	$-t$
Back	$L \times H$	out, L fingers	out, H fingers	XZ	$-W$
Left	$W \times H$	out, W fingers	in, H fingers	YZ	0
Right	$W \times H$	out, W fingers	in, H fingers	YZ	$L - t$

The XZ plane faces $-Y$, so a negative offset moves the front and back sketches toward $+Y$. Each panel is extruded by t along its sketch's normal, so it lands inside the box.

Step 1: Open the ParamWeave workbench and the graph pane

- 1 Install the workbench once with `python3 tools/install_dev.py` and restart FreeCAD.
- 2 Pick **ParamWeave** in the workbench selector.
- 3 Create a new document (**File** → **New**).
- 4 If the graph pane isn't showing, click **Toggle Graph**.

The pane on the right is the graph canvas; its lower half is the property panel. The status line reads *0 nodes, 0 wires*.



The ParamWeave workbench with an empty graph pane beside the 3D view

Step 2: Add the constants

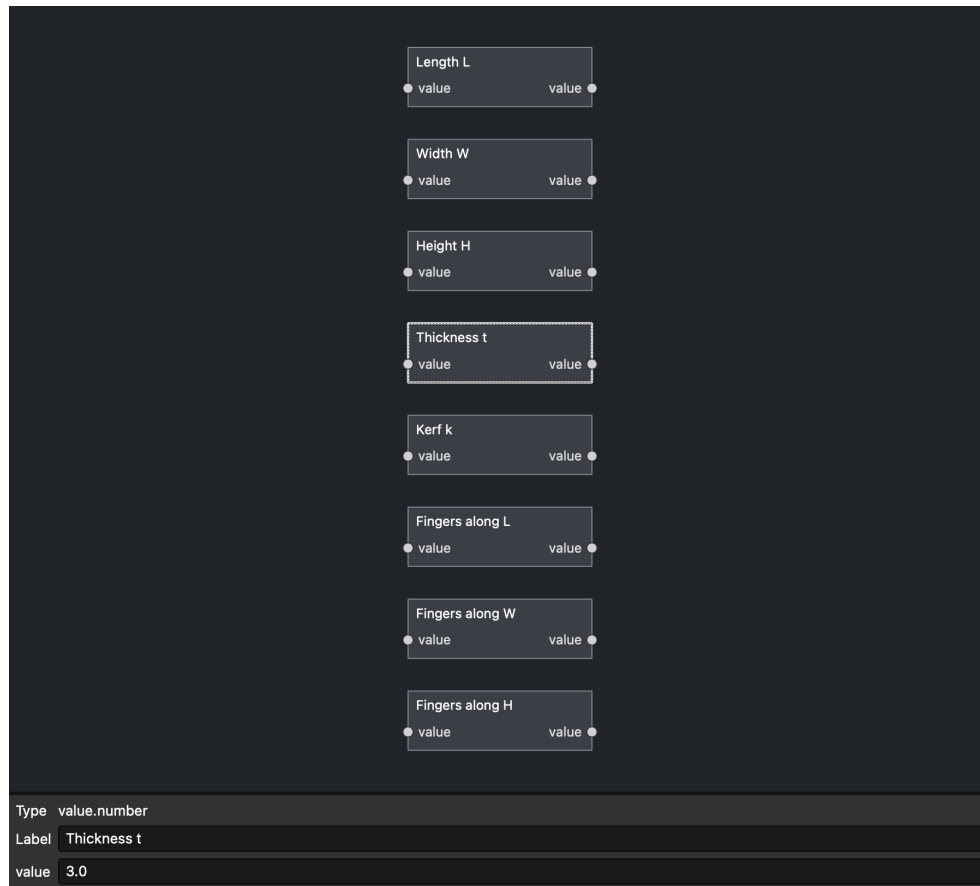
Add eight **Values** → **Number** nodes, one for each constant. For each one, set **Label** and **value** in the property panel:

Label	value
Length L	160
Width W	100
Height H	70
Thickness t	3
Kerf k	0
Fingers along L	7
Fingers along W	5

Label	value
Fingers along H	3

Stack them in a column at the left of the canvas. These are the only numbers you will ever need to change.

Kerf k is the width of material your laser or cutter burns away. Leave it at 0 while modelling. Before cutting, set it to your machine's kerf (often 0.1–0.2 mm): every panel outline then grows by half the kerf, so the fingers and slots come out at their nominal size and the joints fit tight.



Eight Number nodes; Thickness t is selected and its value is shown in the property panel

Step 3: Add the expressions

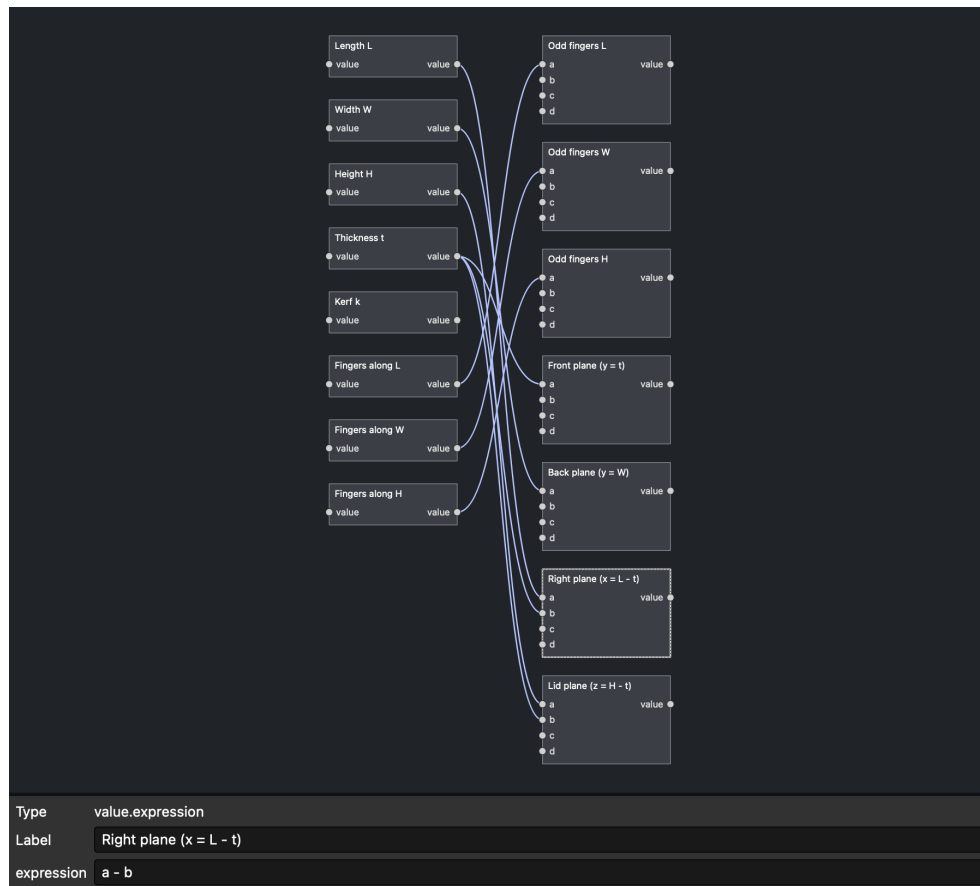
Add seven **Values** → **Expression** nodes in a second column. An Expression computes its **expression** text from its inputs **a**, **b**, **c** and **d**. Set the Label and expression, then wire the listed constants' **value** outputs into the listed inputs:

Label	expression	Wires in
Odd fingers L	$\max(1, 2 * \text{floor}(a / 2) + 1)$	Fingers along L → a
Odd fingers W	$\max(1, 2 * \text{floor}(a / 2) + 1)$	Fingers along W → a
Odd fingers H	$\max(1, 2 * \text{floor}(a / 2) + 1)$	Fingers along H → a
Front plane (y = t)	-a	Thickness t → a

Label	expression	Wires in
Back plane ($y = W$)	$-a$	Width $W \rightarrow a$
Right plane ($x = L - t$)	$a - b$	Length $L \rightarrow a$, Thickness $t \rightarrow b$
Lid plane ($z = H - t$)	$a - b$	Height $H \rightarrow a$, Thickness $t \rightarrow b$

The three "Odd fingers" expressions round any count up to the next odd number, so typing 8 fingers gives 9 instead of an error. The other four compute where each panel's sketch plane sits.

Expressions support $+$, $-$, $\times$, $\div$, $**$, $\min$, $\max$, floor , ceil , round , sqrt , abs , clamp , degree-based trig ($\sin$, $\cos$, atan2 , ...), π , and x if condition else y .

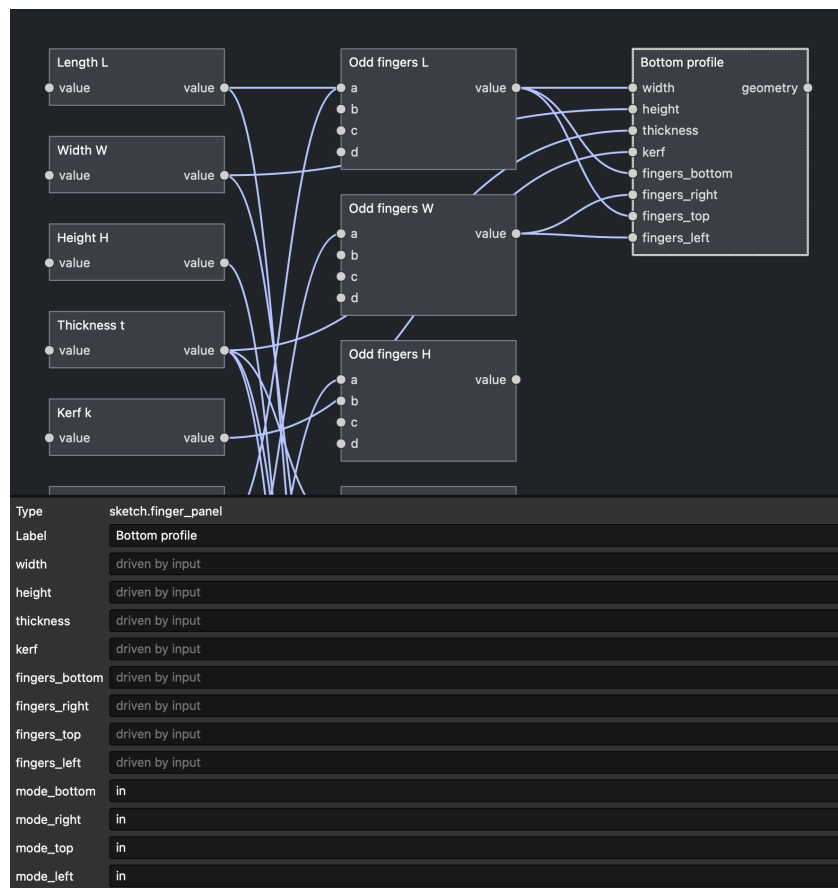


Expressions wired to the constants; the Right plane expression uses two inputs

Step 4: Add the bottom panel profile

- 1 Add **Sketch Elements** → **Finger Joint Panel** in a third column and label it **Bottom profile**.
- 2 Set all four modes, **mode_bottom**, **mode_right**, **mode_top** and **mode_left**, to **in**.
- 3 Wire the panel's size and finger counts:
 - Length L → **width**, Width W → **height**, Thickness t → **thickness**, Kerf k → **kerf**
 - Odd fingers L → **fingers_bottom** and **fingers_top**
 - Odd fingers W → **fingers_right** and **fingers_left**

The wired fields now read *driven by input*. Because every other panel starts as a duplicate of this chain, they all inherit the thickness and kerf wires. The edge modes are plain text fields further down the property panel.

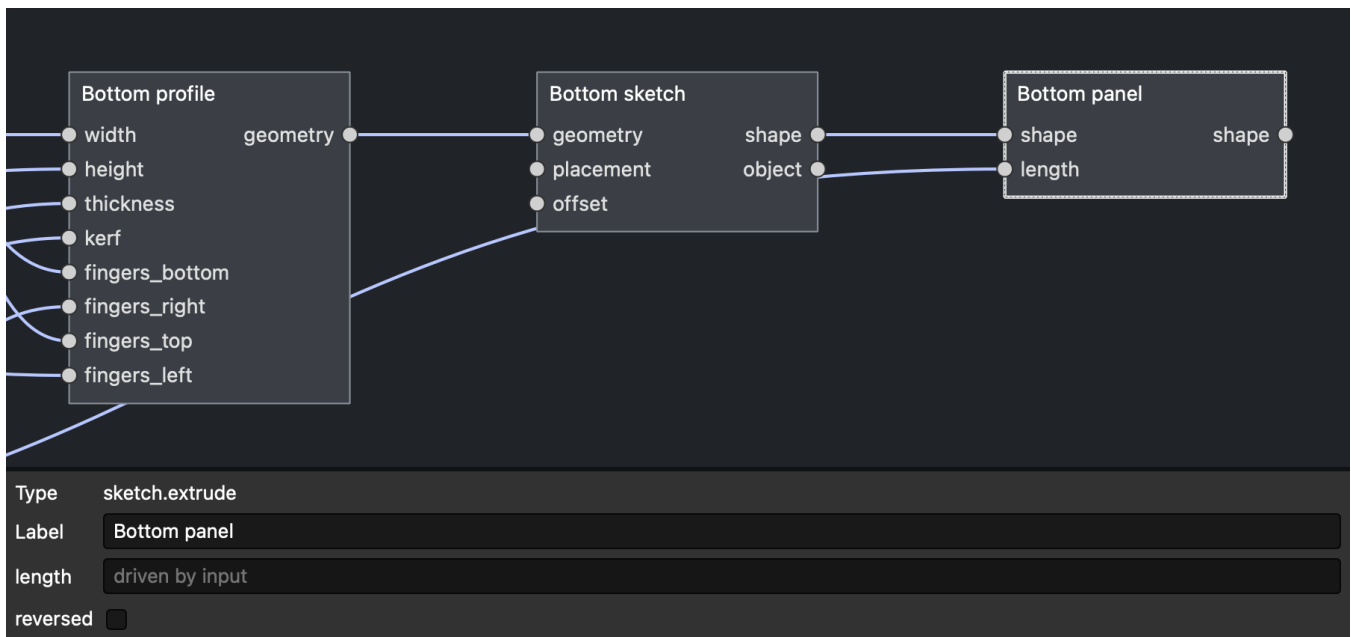


The bottom profile wired to the constants; the property panel scrolled to its edge modes

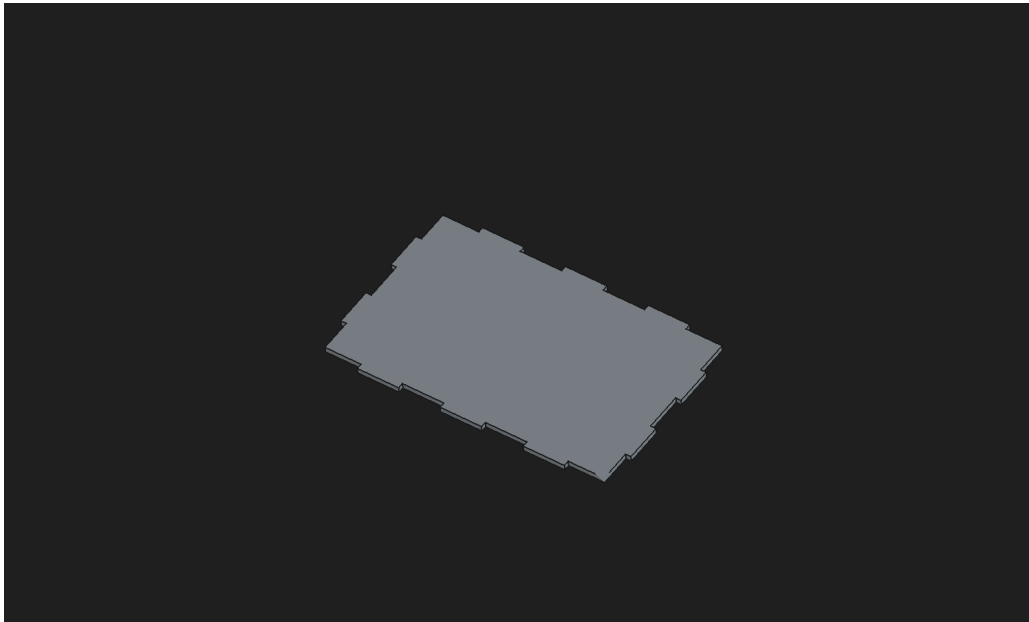
Step 5: Turn the profile into a sketch and extrude it

- 1 Add **Sketch** → **Sketch**, label it `Bottom sketch`, and wire *Bottom profile geometry* → **geometry**. Leave **plane** at `XY` and **offset** at 0.
- 2 Add **Construct** → **Extrude**, label it `Bottom panel`, and wire *Bottom sketch shape* → **shape**. Then wire Thickness `t` → **length**.
- 3 Click **Evaluate**.

Every node turns green (*ok*). FreeCAD now has a real **Bottom sketch** (a Sketcher object you can open) and a **Bottom panel** solid in the *ParamWeave Generated* group. The sketch is hidden because the extrude uses it.



Profile, Sketch and Extrude chain for the bottom panel

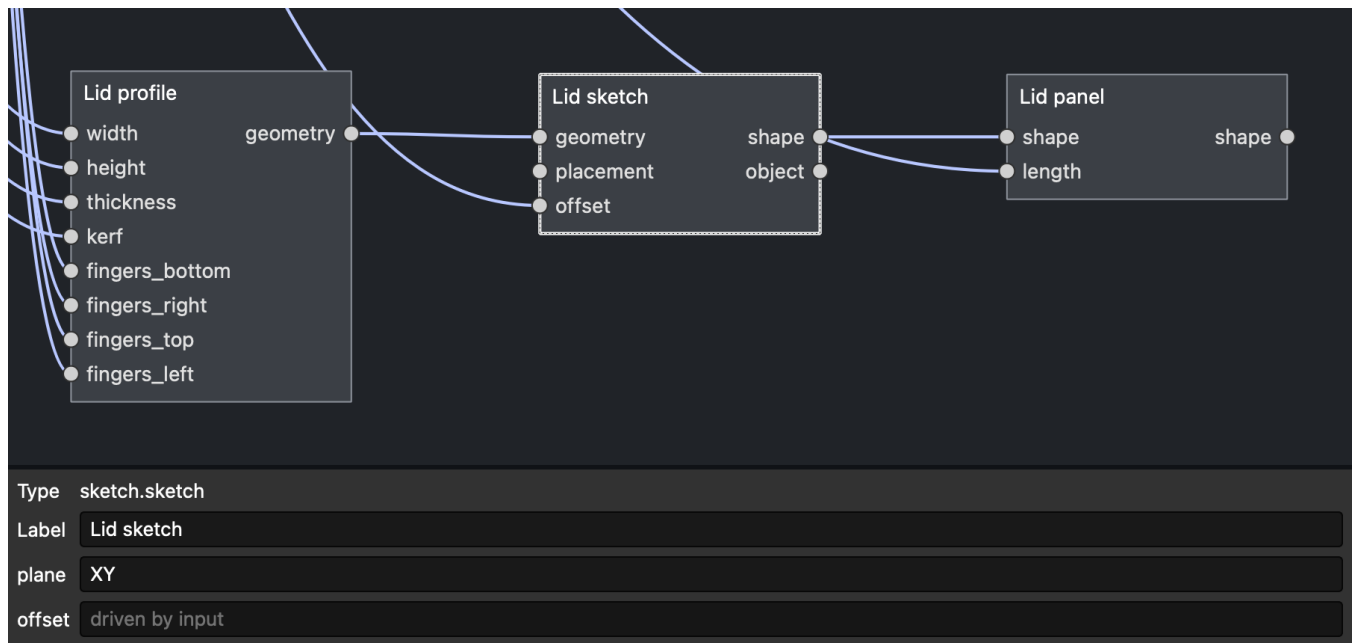


The bottom panel: slots on all four edges

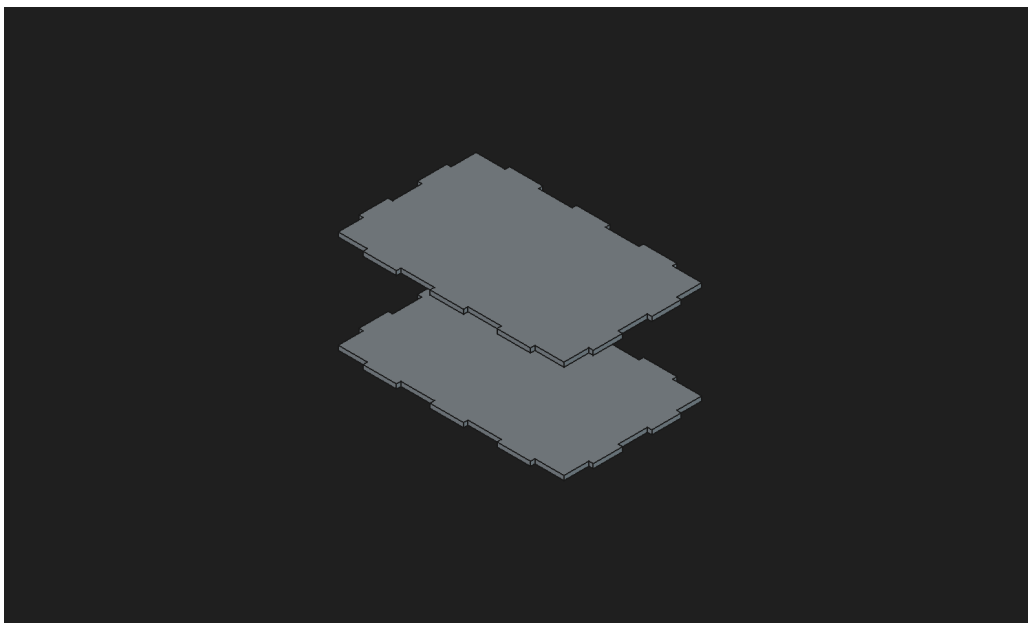
Step 6: Duplicate the bottom chain to make the lid

The lid is identical to the bottom, just raised to the top.

- 1 Click **Bottom profile**, then Ctrl-click (⌘-click) **Bottom sketch** and **Bottom panel**.
- 2 Press **Ctrl+D** (⌘D). Three copies appear, already wired to the same constants, and they are selected.
- 3 Drag the copies below the other panels and rename them **Lid profile**, **Lid sketch** and **Lid panel**.
- 4 Wire the **Lid plane ($z = H - t$)** expression's **value** → **Lid sketch offset**.
- 5 Click **Evaluate**.



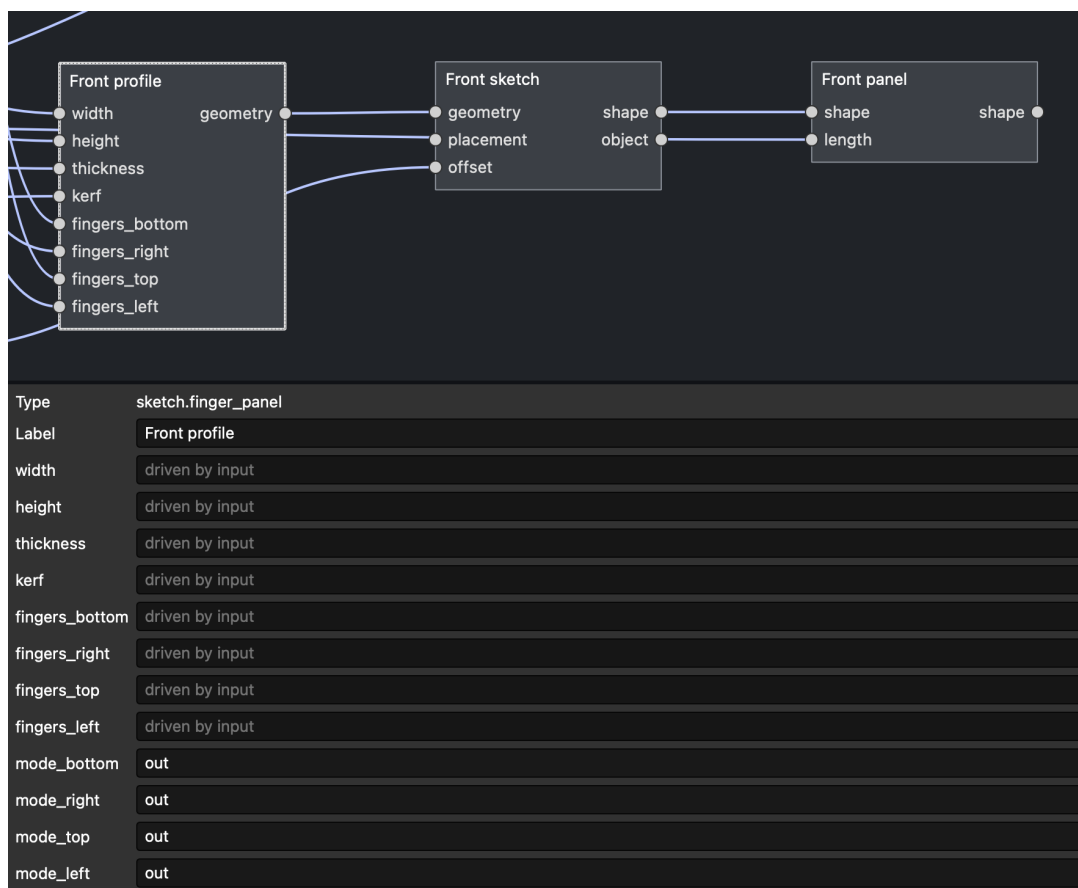
The lid chain, a copy of the bottom chain with its sketch offset wired to the lid plane



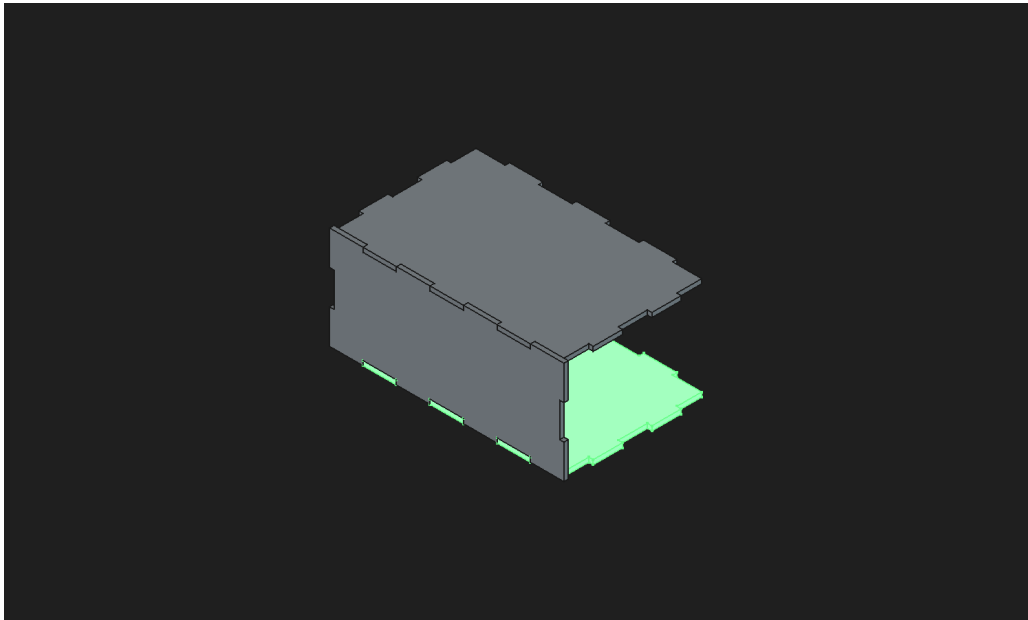
Bottom and lid panels

Step 7: Duplicate again for the front wall and change it

- 1 Select the three **bottom** nodes again, duplicate them, drag the copies into a new row and rename them **Front profile**, **Front sketch** and **Front panel**.
- 2 On **Front profile**, set all four modes to **out**.
- 3 Re-wire the front profile's height and side finger counts. Wiring into an occupied input replaces the old wire:
 - Height H → **height** (replaces Width W)
 - Odd fingers H → **fingers_right** and **fingers_left**
- 4 On **Front sketch**, set **plane** to **XZ** and wire **Front plane (y = t)** → **offset**.
- 5 Click **Evaluate**.



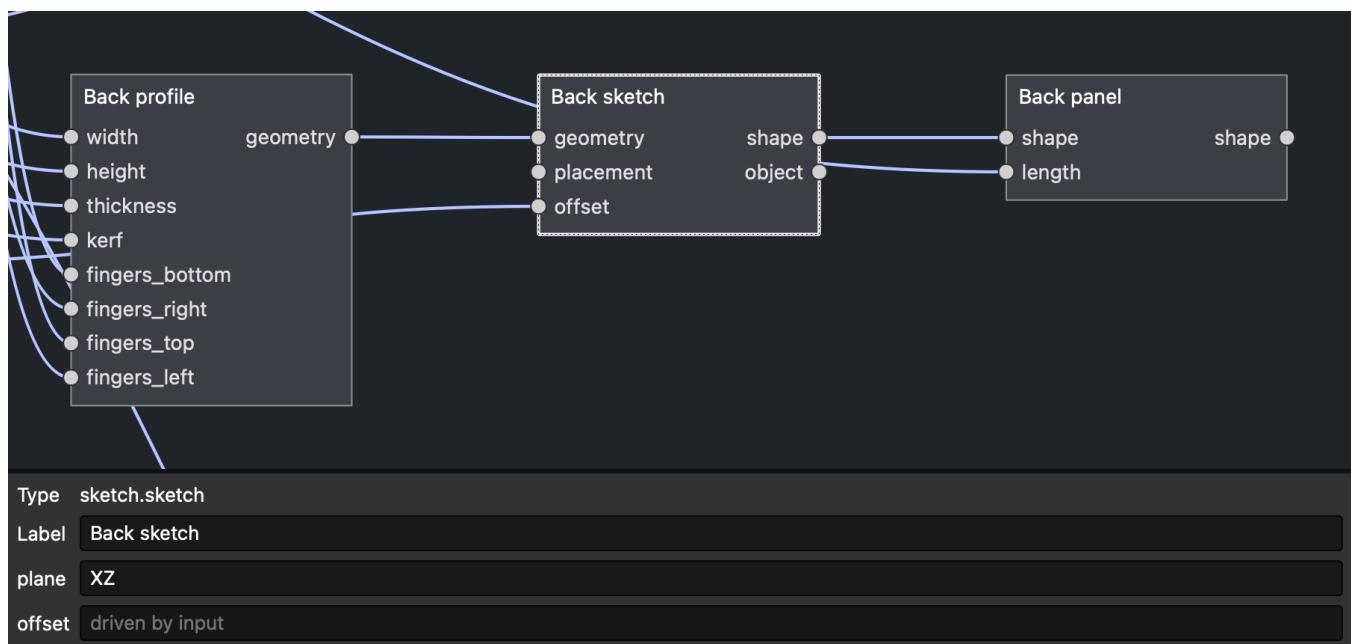
Front chain: height and side finger counts re-wired, all edge modes set to out



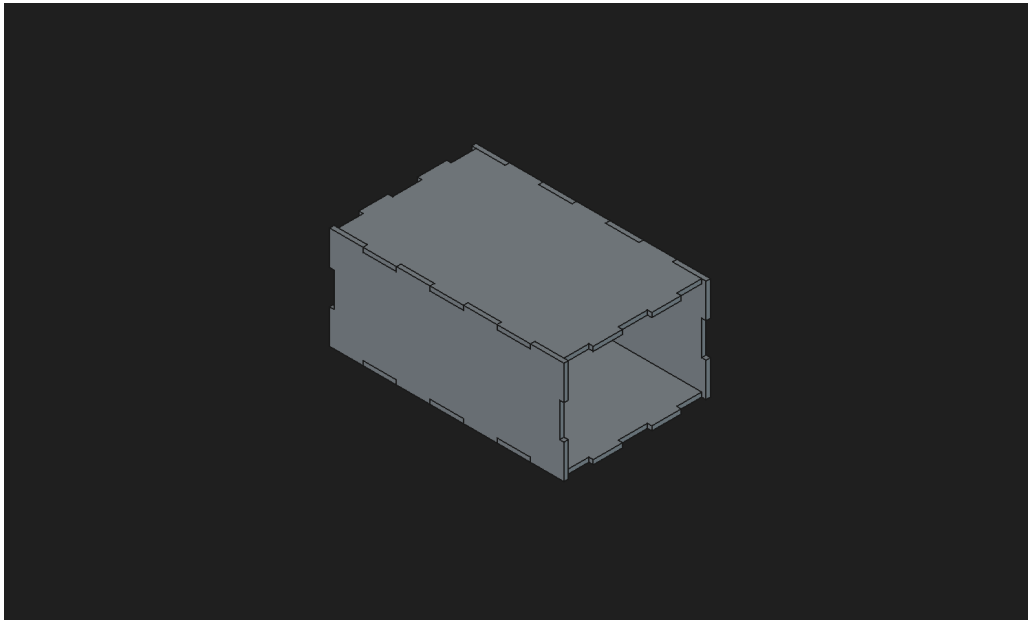
The front wall's fingers interlock with the slots in the bottom and lid

Step 8: Duplicate the front wall to make the back wall

- 1 Select the three **front** nodes, duplicate them, move the copies into a new row and rename them **Back profile**, **Back sketch** and **Back panel**.
- 2 Wire **Back plane (y = W)** → **Back sketch offset** (replaces the front plane wire).
- 3 Click **Evaluate**.



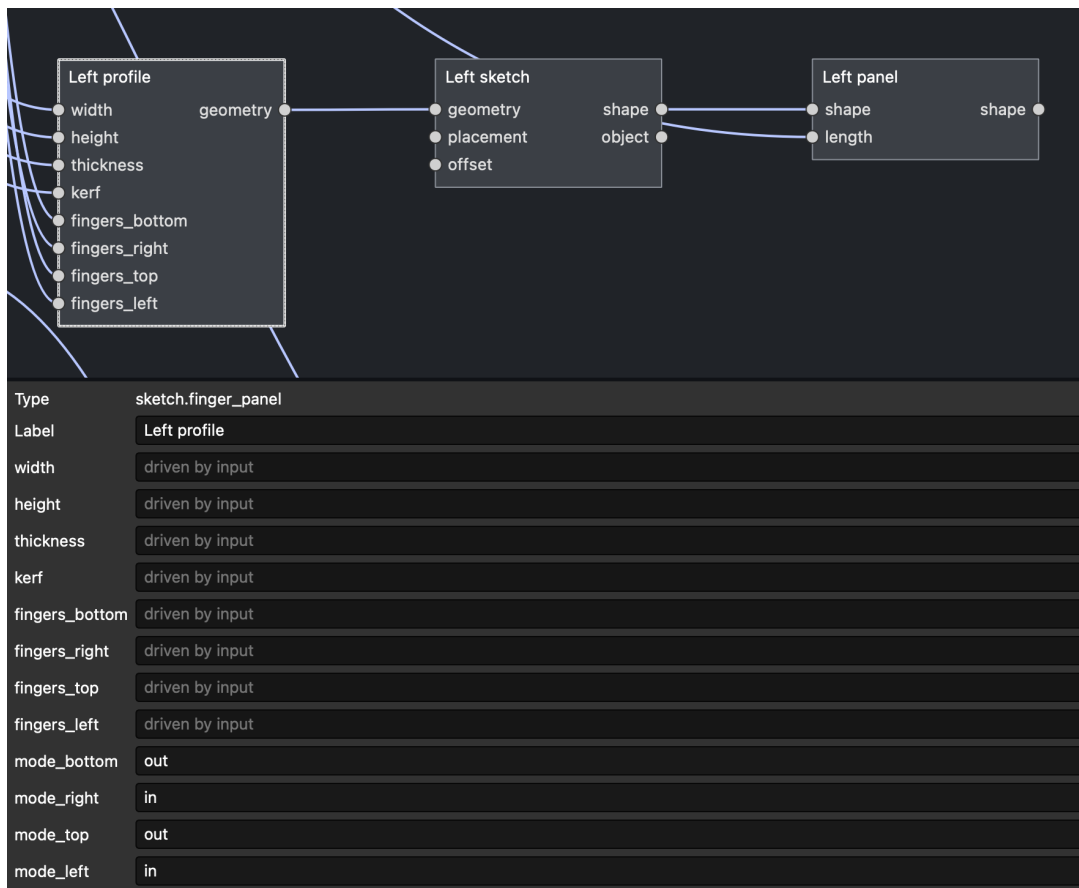
The back chain differs from the front only in its sketch offset



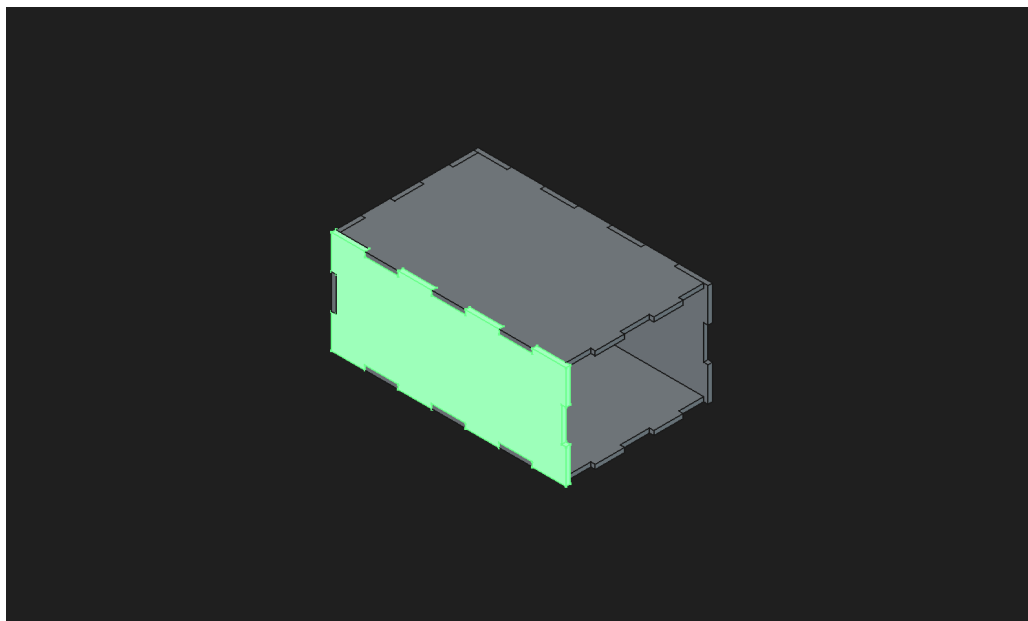
Front and back walls in place

Step 9: Duplicate the front wall for the left wall and change it

- 1 Duplicate the three **front** nodes once more. Move the copies into a new row and rename them **Left profile**, **Left sketch** and **Left panel**.
- 2 On **Left profile**:
 - wire Width W → **width**;
 - wire Odd fingers W → **fingers_bottom** and **fingers_top**;
 - set **mode_right** and **mode_left** to **in**, so the vertical edges take the front and back walls' fingers.
- 3 On **Left sketch**, set **plane** to **YZ**.
- 4 The left wall sits at $x = 0$, so it needs no offset wire. Click the wire going into **Left sketch offset**, press **Delete**, then set **offset** to **0**.
- 5 Click **Evaluate**.



Left chain: YZ plane, width re-wired to W, vertical edges switched to in

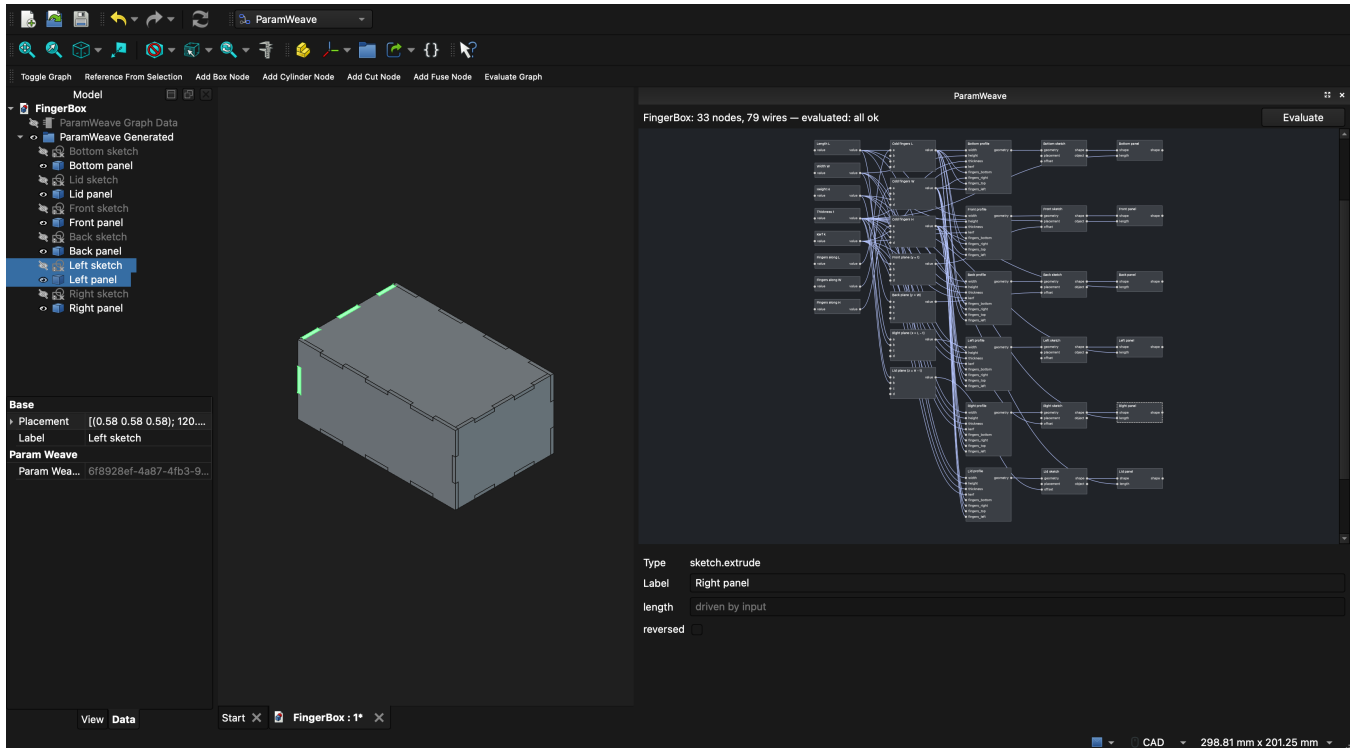


The left wall (highlighted) slots between the front and back walls

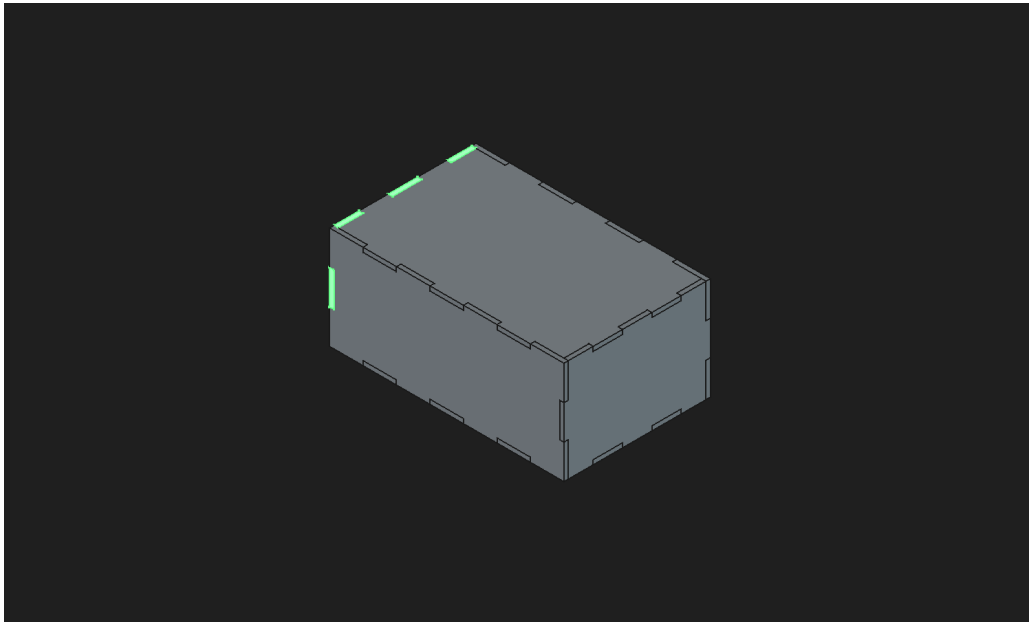
Step 10: Duplicate the left wall to make the right wall

- 1 Duplicate the three **left** nodes, move them into the last row and rename them **Right profile**, **Right sketch** and **Right panel**.
- 2 Wire **Right plane (x = L - t)** → **Right sketch offset**.
- 3 Click **Evaluate**, then press **F** to frame the whole graph.

The status line reads *33 nodes, 79 wires — evaluated: all ok*. The model tree lists six hidden sketches and six visible panels.



The complete graph, the generated objects in the model tree, and the assembled box



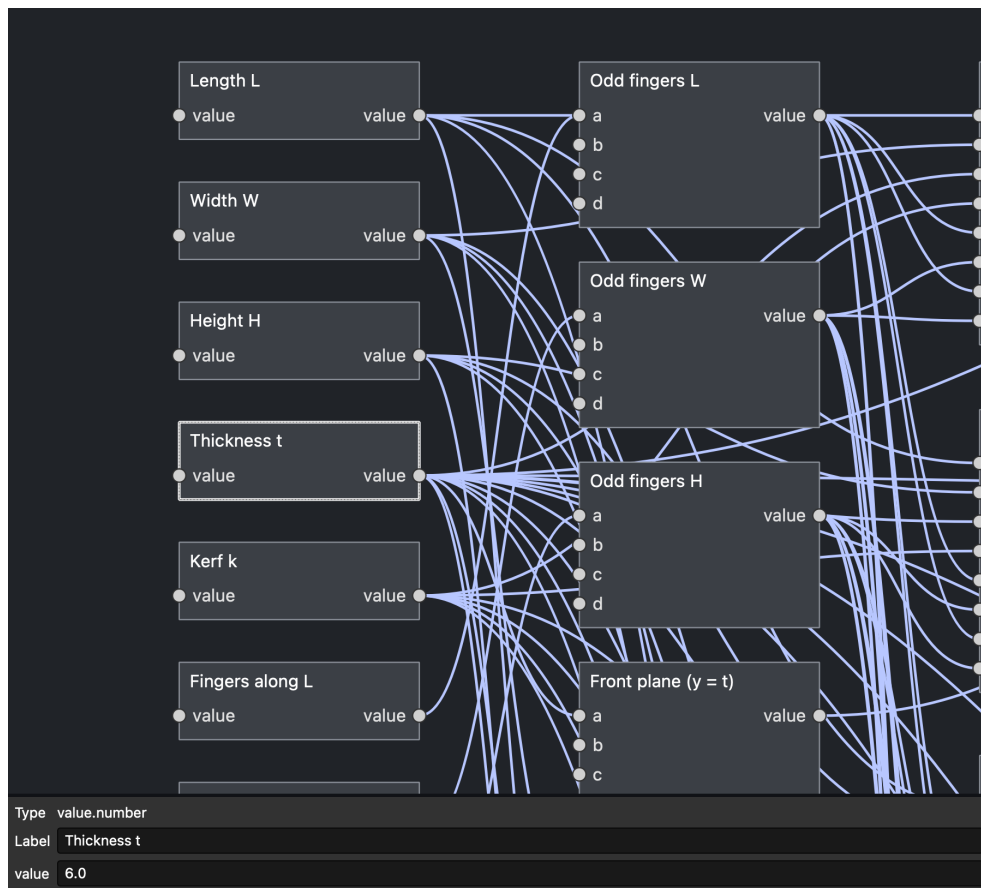
The finished finger-jointed box

Step 11: Change a constant and re-evaluate

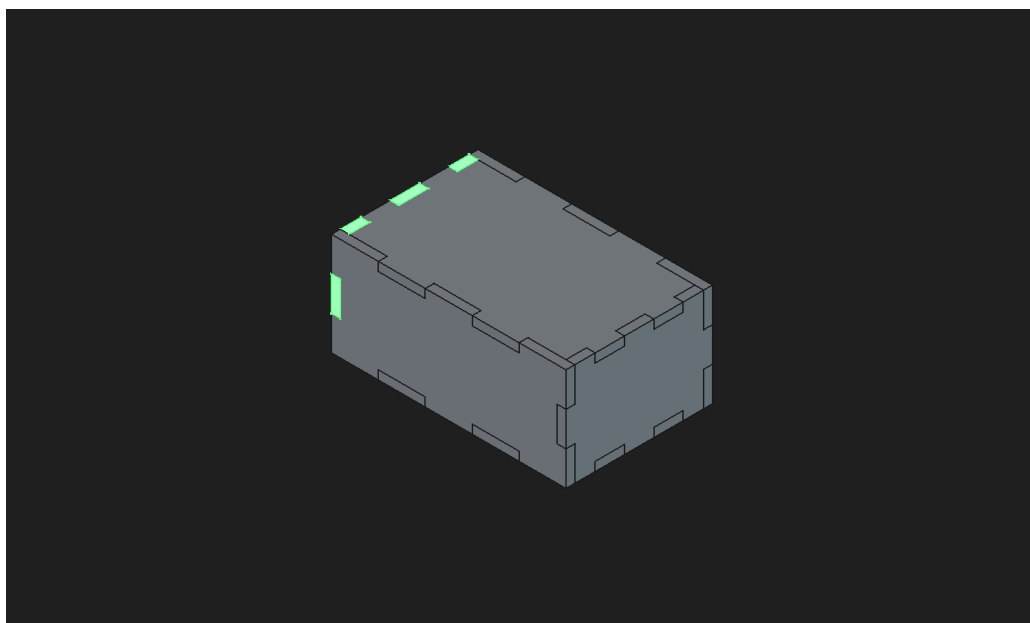
This is where the graph pays off. Select **Thickness t** and set it to 6, then select **Fingers along L** and set it to 4. Click **Evaluate**.

All six panels regenerate in place, without duplicating any objects:

- every slot is now 6 mm deep;
- the length edges show 5 fingers (4 rounded up to the next odd number);
- the lid and right-wall planes move inward to follow the new thickness.



Thickness raised to 6 and fingers along L set to 4 (rounded up to 5)



The same box regenerated with 6 mm material and five fingers along its length

Where to go next

- **Export for cutting:** choose **ParamWeave** → **Export Cut Files...** (or right-click the canvas). Pick a sheet width and gap, then save a **DXF** or **SVG**: all six panels are laid flat on one sheet, with the kerf applied and construction lines left out. The SVG uses red hairline strokes, which most laser software treats as cut lines.
- **Open-top box:** delete the three lid nodes and set **mode_top** to `flat` on the four wall profiles.
- **Drive from a spreadsheet:** replace a Number node with **Values** → **Document Variable** that points at a Spreadsheet alias.
- **Cut it:** set **Kerf k** to your cutter's kerf (for example 0.15) before exporting; Export Cut Files evaluates the graph for you. The 3D panels then overlap slightly, which is expected: the sketches are now cutting outlines, grown by half the kerf.
- **Organise with frames:** select a panel's three nodes and press **Ctrl+G** (⌘G) to wrap them in a frame; give it a label and colour in the property panel. Dragging the frame's title bar moves the whole panel chain, and Ctrl+D on a frame copies the chain inside it.
- **Shortcut:** **ParamWeave** → **Example: Finger-Jointed Box** inserts this same graph in one step, already organised into frames.